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Class 9
Biology
Chapter 6

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Unit no. 6:

Topic: Molecular Biology

6.1: Biochemistry/Molecular Biology

1: Define Biochemistry.

Biochemistry: Biochemistry is the study of different chemical compounds and the chemical processes taking place within the living organisms.

2. What is the importance of Biochemistry?

Ans: Study of Biochemistry is greatly helpful to explore the cell biology and anatomy as all the structures of living organisms have specific biochemical organization and it is also essential to study the physiology of organisms.

3. Define bio elements?

Any chemical element that is found in the molecules and compounds that make up living organisms are called bio elements. Out of 92 naturally occurring elements of earth, 16 of these bio elements.

Types of bio elements: bio elements are classified into two categories.

- Major bio elements: Six most common bio elements constitute 99% of protoplasm so called major bio elements. They are Oxygen, Carbon, Hydrogen, Nitrogen, Calcium, Phosphorus.
- Minor bio elements: Remaining ten bio elements constitute only 1% of the biomolecules so called minor bio elements. They are potassium, sulphur, chlorine, sodium, magnesium, iron, copper, manganese, zinc and iodine.

4. Compare ratios of bio-elements.

Ratio of bio-elements:

- Oxygen 65%
- Carbon 18%
- Hydrogen 10%
- Nitrogen 3%
- Calcium 2%
- Phosphorus 1%



6.2 Biological Molecules

1. Compare the energy level of carbohydrates, proteins, lipids, DNA and RNA.

Biomolecules	Examples	Location in cell	Energy level	Main Role
Carbohydrates	Glucose, lactose, starch, cellulose	Cell wall, cell membrane, smooth Endoplasmic reticulum, Golgi Apparatus	4 kcal/g	Used as food, energy source, component of cell wall, DNA and RNA
Proteins	Hemoglobin, hormones, enzymes, antibodies	Cytoplasm, rough endoplasmic reticulum, golgi apparatus	4 kcal/g	Energy source, component of cell membrane, gaseous transport, chemical messenger, speed up chemical reaction, immunity
Lipids	Fats, waxes, oils, vitamin D	Cell membrane, smooth endoplasmic reticulum	9 kcal/g	Provide cells with long term energy, component of cell membrane
DNA		Nucleus (chromosomes)		Component of chromosome, store and pass on genetic information
RNA	mRNA, tRNA, rRNA	Nucleolus, cytoplasm		Tools for translation, protein synthesis

Topic 6.3: Protein

1. Describe Structure, Function and Source of Protein.



Protein:

Proteins are main structural and functional components of cell, which make up around 55% of the dry weight of cell. Protein are synthesized by Ribosomes.

Chemical Composition:

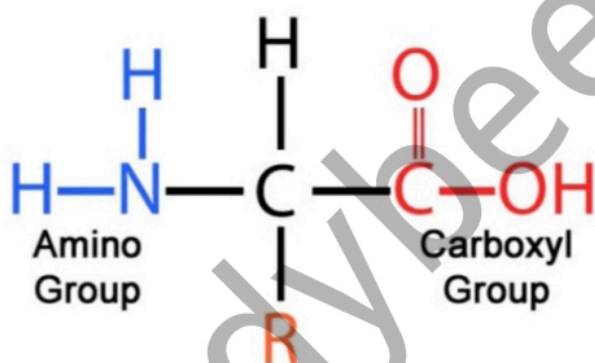
It is composed of C, H, O, N and sometimes P, S, Fe etc. Chemically Proteins are polypeptide chains which are polymers of Amino acids.

Amino acids:

Amino acids are the building blocks of proteins. About 20-25 different types of Amino acids are found in protein.

Structure of Amino acid:

Amino acids contain a (alpha) carbon. Four different groups are attached to a-carbon, including a Hydrogen atom (-H), an amino group (-NH₂), a carboxyl group (-COOH) and an alkyl group (-R). Each amino acid has different -R group which determines its chemical properties.



Structure of Protein:

Structure of proteins may consist of a single polypeptide or more than one polypeptide chain. Number of Amino acids and peptide chain varies in different proteins.

Peptide bond:

Bond b/w two amino acids is called peptide bond.

Types of proteins:

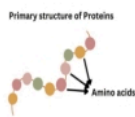
Protein is classified into four categories.



- Primary Proteins
- Secondary
- Tertiary
- Quaternary

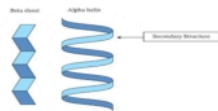
Primary Proteins:

A linear polypeptide with a specific number, type and sequence of amino acids is called primary Proteins.



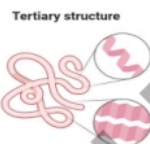
Secondary proteins:

In secondary proteins, there is folding of polypeptide with extra bonds.



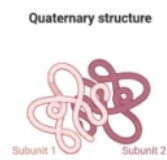
Tertiary proteins:

In tertiary proteins, there is further folding of secondary structure with even extra bonds.



Quaternary proteins:

Two or more polypeptide with tertiary folding bonded with each other is called quaternary proteins.



Examples and Function of proteins



Examples	Functions
Cell membrane proteins	Act as channel and antigen
Enzymes	Control biochemical reactions
Hormones	Act as chemical messenger and regulate different functions in body like growth, reproduction etc
Antibodies	Provide immunity against germs
Fibrinogen	Blood clotting protein
Actin and Myosin	Involved in contraction and relaxation of muscles
Collagen	Provide support and strength
Keratin	Present in hair, nail, feathers, beak and make them hard and protective
Histone	Attached to DNA to form protein
Haemoglobin	Transport gases in body via blood

Sources of protein:

We obtain protein from:

- Plants seeds like beans, lentils, peas, nuts
- Milk
- Beef
- Mutton
- Egg
- Fish
- Sea food
- Mushrooms

Topic 6.4: Lipids

Describe Structure, Function and source of Lipids.

Lipids:



Lipids are heterogeneous group of organic compounds which are insoluble in water but soluble in organic solvent such as alcohol, ether etc

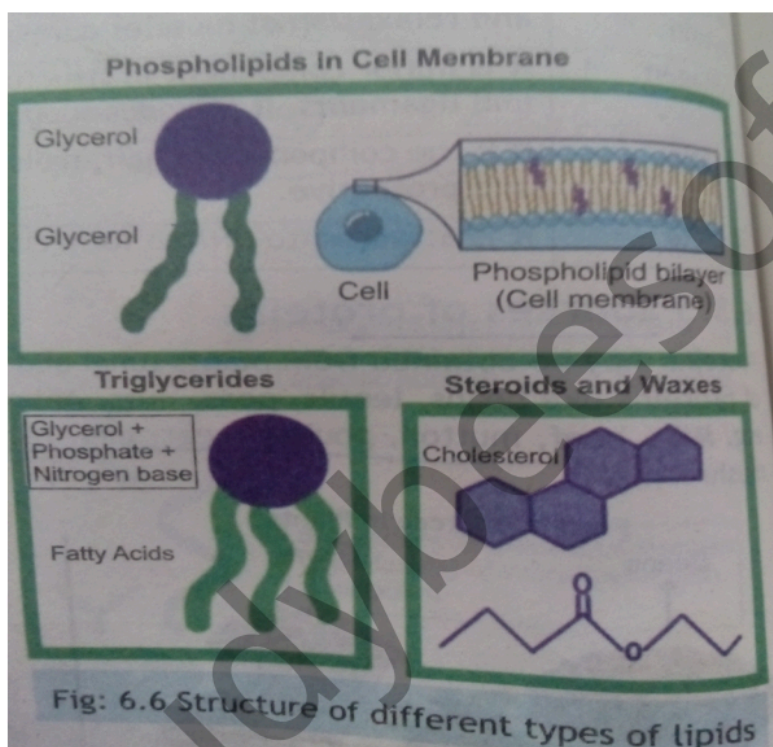
Composition:

Lipids are composed of C, H and O. One gram of lipids contains 9kcal energy.

Structure of lipids:

Lipids are made up of three fatty acids and a glycerol molecule are called tri-acyl glyceride.

Glycerol + 3 fatty acids $\longrightarrow$ Tri-acyl glyceride



Types of fatty acids:

Fatty acids are classified into two types:

- Saturated fatty acids
- Unsaturated fatty acids

Saturated fatty acids:

They have all C-C single bonds and solid at room temperature. e.g cheese, butter, ghee etc



Unsaturated fatty acids:

They have double or triple C-C bonds and are liquid at room temperature. e.g soybean oil, mustard oil, olive oil etc.

Functions of lipids:

- Lipids are components of plasma membrane.
- They act as energy stores in fat cells, liver and in blood.
- They act as important steroid hormones e.g testosterone, aldosterone etc
- They also act as insulators against atmospheric heat and cold.
- Some vitamins are lipid in nature. e.g Vitamin D is derived from cholesterol.

Sources of lipids:

We obtain lipids from:

- Red meat
- Butter
- Cheese
- Ghee
- Fish liver oil
- Sunflower oil, soybean oil, peanut and coconut oil
- Waxes are obtained from comb of many insects.

Topic 6.5: Carbohydrates

Q.1: Describe the structure, function and sources of Carbohydrates.

Carbohydrates:

Carbohydrates are the "hydrates of carbon". They are also called as sugars or saccharides due to their sweet nature. Word saccharide is derived from Greek word "Sakcharon" meaning sugar.

Structure and energy producing nature of Carbohydrates:

Carbohydrates are composed of carbon, hydrogen and oxygen with the ratio of 1:2:1. Hydrates of carbon name is given because number of carbon atoms are equal to the number of water molecules.



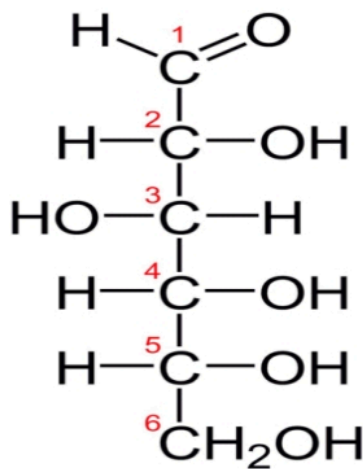
Glucose is the simplest form of carbohydrates which is naturally produced by the process of photosynthesis. One gram of glucose contains about 4kcal energy.

General formula of Carbohydrates:

The general formula of Carbohydrates is $C_x(H_2O)_y$ where X is number of carbon atom and Y is number of water molecules.

For example:

The formula of glucose is $C_6H_{12}O_6$.



Structure of glucose

Functions of carbohydrates:

- Carbohydrates are the major source of energy.
- Carbohydrates are structural material for cell wall.
- Cellulose present in the cell wall of fruits help to avoid constipation.
- It also helps in lowering blood cholesterol level.
- It regulates blood pressure.
- Sucrose is used as common table sugar.
- Extra blood glucose is stored in the form of glycogen.
- Chitin makes exoskeleton of arthropods.



Sources:

The sources of Carbohydrates are:

- Fruits
- Vegetables
- Dairy products
- Wheat
- Barley
- Rice
- Potato
- Cotton which is pure cellulose

Topic 6.6: Classification of Carbohydrates

Q.1: Write the classification of Carbohydrates.

Classification of Carbohydrates:

Carbohydrates are classified into three groups:

- Monosaccharide
- Disaccharide
- Polysaccharide

Monosaccharide:

- Monosaccharides are the simplest form.
- They are produced during photosynthesis.
- They are not hydrolysed.
- They are white crystalline solid.
- They are sweet in taste.
- They are soluble in water.
- Range of C-atom is 3-7



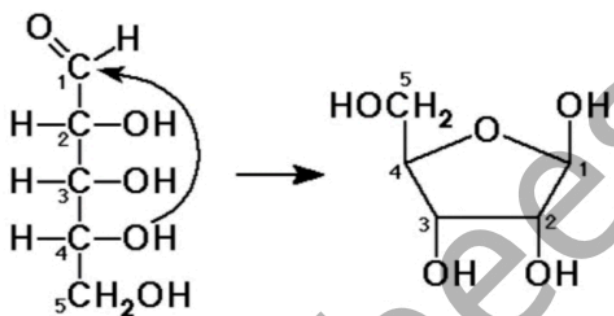
Name of monosaccharide:

Monosaccharides are named on the basis of number of carbon:

- Trioses (3C)
- Tetraoses (4C)
- Pentoses (5C)
- Hexoses (6C)

Common monosaccharide:

Common monosaccharide are Pentoses and Hexoses. Ribose is a pentose found in the nucleotide of nucleic acid and Glucose is hexose.



Open structure Ribose → Ring structure Ribose

Disaccharides and their formation:

- Disaccharides are formed due to bonding of two monosaccharides.
- They can be hydrolysed.
- They are white crystalline solid.
- They are less sweet in taste.
- They are less soluble in water.
- Their molecular formula is $C_{12}H_{22}O_{11}$

Types of Disaccharides and their sources:

Types of Disaccharides are:

- Sucrose



- Maltose
- Lactose

Sucrose:

Sucrose is formed by the condensation of glucose and fructose releasing a water molecule. It is found in sugar cane.



Maltose:

Maltose is formed by the condensation of two glucose molecules releasing a water molecule. Malt sugar is found in cereals, beer and related drinks.



Lactose:

Lactose is formed by the condensation of galactose and glucose. It is found in milk and milk products.



Polysaccharide:

- Polysaccharide is the largest group of Carbohydrates.
- They are complex and polymers of more than 10 monosaccharides.
- They are storage molecule of carbohydrates.
- They are not sweet in taste.
- They can be hydrolysed.
- They are the structural component of cell wall of plant, fungi and bacteria.
- Examples are starch, glycogen, cellulose

Topic 6.7: Nucleic Acids

Q.1: Describe briefly the structure of DNA as a double helix macromolecules made of



nucleotides with base pairing in between the two helices through complementary base pairing.

Nucleic Acids:

Swiss physician Friedrich Miescher isolated a whitish material from nucleus of human pus cells. He named this material as nuclein as it was isolated from nucleus. Later on, it was named as nucleic acid because of its acidic nature.

Function of Nucleic Acids:

Nucleic Acids are organic molecules which store and transfer hereditary information from one generation to next.

Types:

There are two types of nucleic acid:

- Deoxyribonucleic Acid (DNA)
- Ribonucleic Acid (RNA)

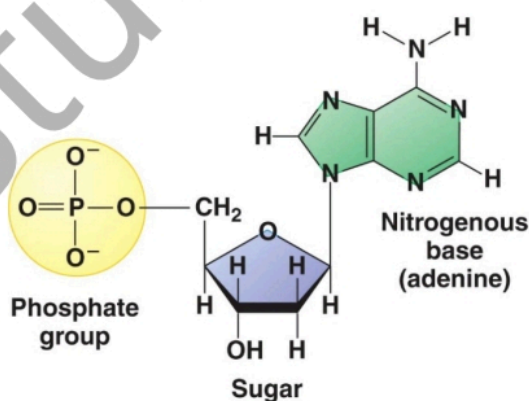
Nucleotides:

DNA and RNA are made up of many units called nucleotides.

Structure of Nucleotide

Nucleotide is made up of three components.

1. Pentose (5C sugar)
2. Nitrogenous bases which is attached at the first carbon of pentose sugar.
3. Phosphate group Acid attached with fifth carbon of pentose sugar.



Types of pentose sugar:

- Deoxyribose in DNA
- Ribose in RNA

Types of nitrogen bases in nucleic acid:

There are two types:

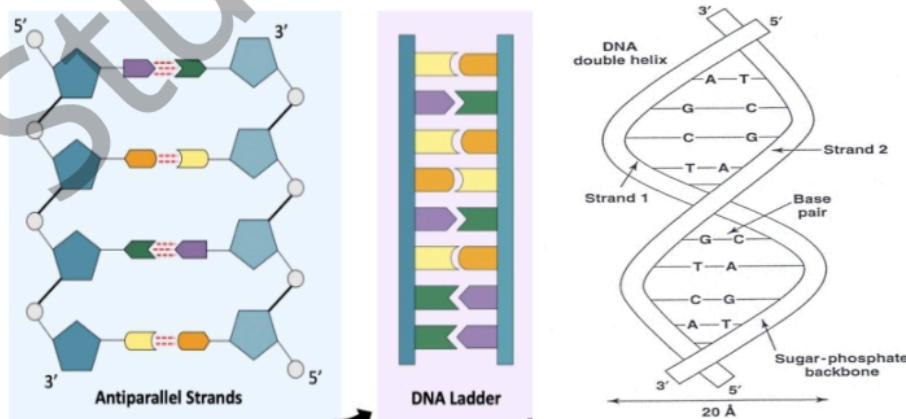
- Purines, which are double ring structure. e.g Adenine and Guanine
- Pyrimidines which are single ring structure i.e. Cytosine, Thymine and Uracil

Difference b/w DNA and RNA

DNA	RNA
Sugar is deoxyribose	Sugar is ribose
Adenine, Thymine, Cytosine and Guanine are present.	Adenine, Uracil, Cytosine and Guanine is present.
Double stranded molecule	Single stranded molecule
Transfer genetic information	Synthesis protein
Present in nucleus	Present in Nucleus and Cytoplasm.

Structure of DNA:

Structure of DNA is explained by two scientists, James Watson and Francis Crick. DNA molecule is a polynucleotide in which phosphate group is attached to C-5, nitrogen base is attached to C-1, OH is attached to C-3 and H is attached to C-2 of pentose sugar.



The main points of Watson and Crick Model of DNA are:

- The DNA molecule is double stranded like ladder and both strands are coiled around each other like a helix.
- Both strands are antiparallel.
- The back bone of molecule are composed of pentose sugar, phosphate group and nitrogenous bases.
- The amount of Adenine and thymine is always equal and amount of Cytosine and Guanine is always equal in DNA.
- Adenine always pair with Thymine and Cytosine always pair with guanine.
- There are two hydrogen bonds b/w Adenine and thymine and three hydrogen bond b/w Cytosine and Guanine.
- The diameter of double helix of DNA is about 2nm.

Topic 6.8: Function of DNA (genes)

Q.1 Define Genes?

Genes:

The basic unit of heredity which is transferred from a parent to offspring and is held to determine some characteristics of the off spring. Gene is a localized region of DNA that codes for a protein.

Q.2 Define Chromatin.

Ans: Hereditary material (DNA) is found in Nucleus in the form of thin fibres called Chromatin.

Q.3 Define Nucleosomes?

Ans: Chromatin is composed of DNA and a proteins called Histones. Coiling of DNA around histones form bead like structures called Nucleosomes.

Q.4 Give an account of genes on Chromosomes.

Chromosomes contain units of heredity: The Genes

During cell division, Chromatin fibres condense and further fold to form thread-like structure called chromosomes. Small piece of huge DNA molecule in the chromosomes is called Genes. Genes actually store and control the heredity information. Each chromosome is made up of large number of genes. The total number of genes found in the complete set of chromosomes of an organism is called genome of that organism.



Q.8 Explain that gene is a unit of inheritance and that it can be copied and passed on to the next generation.

Gamete act as vehicle for transfer of heredity information (genes) to next generation:

During sexual reproduction, gametes or spores are produced by meiosis. Male and female gamete (sperm and egg) contain chromosomes of parent generation. At fertilization, male and female gametes fuse to form zygote, thus genes are transferred to zygote. Zygote develops into the offsprings which are the next generation.

Thus, genes are transferred from one generation to next generation through gametes or spores. So therefore, DNA (genes) in the chromosomes carry the heredity information from one generation to next.

Q. 9 Describe the central dogma stating the role of gene in protein synthesis.

Central dogma of molecular biology:

The flow of biological information, from DNA to RNA and from RNA to protein is known as the "Central dogma of molecular biology" or "Central dogma of life".

Role of Gene in Protein Synthesis:

Protein Synthesis takes place in two steps:

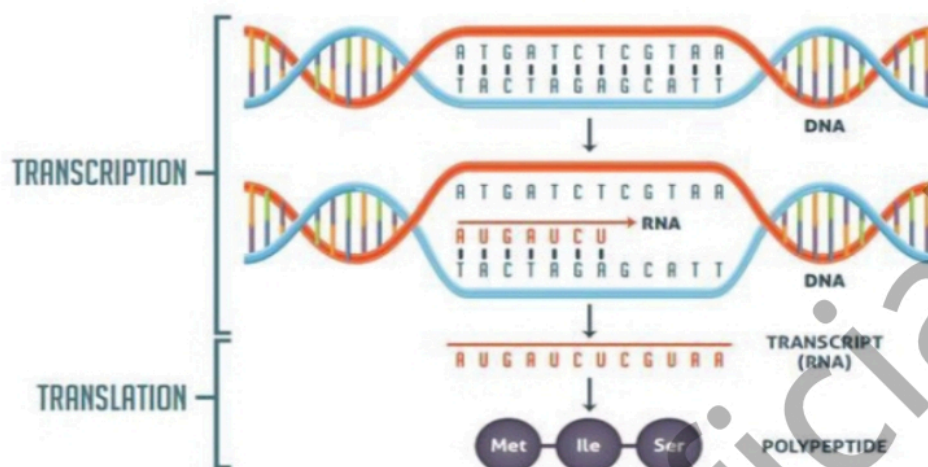
- Transcription
- Translation

Together transcription and translation are known as "gene expression".

Transcription:

Transcription is the first step of gene expression that involves the formation of RNA molecule from DNA. Both Strands of DNA molecule unwind from each other. Only one strand of DNA is copied. Nucleotide sequence of one DNA strand (coding strand) is copied in the form of mRNA. After the gene is copied to mRNA, both strands of DNA are again copied together into its normal double stranded shape.





Transcription and Translation

Translation:

The mRNA synthesized in the nucleus of the cell moves into the cytoplasm where it binds to the ribosome. The mRNA message (codes) is translated by Ribosomes which link amino acids to form polypeptide chain with specific sequence of amino acids. Different types of tRNA transfer amino acids to ribosome as per sequence of codes in mRNA.

Q.10 Explain how genes control inheritance of characters.

The genes control traits by forming a particular protein. Genes are responsible for controlling the traits as they possess the DNA molecules that have the genetic information which is passed to the RNA(Ribonucleic acid) following the transcription process. Then the mRNA (messenger ribonucleic acid) undergoes the translation process to ultimately form the protein. Gene possesses a pair of alleles that helps in instructing the cells to express the traits by making proteins.

The genetic traits are heritable that are passed from the parents to the offspring through deoxyribonucleic acid, which possesses the genetic information. The homologous chromosomes have different traits like eye color, height, and blood type that are passed to the offspring.

Q.11 Explain the types of RNA and their functions?

Ans: All types of RNA are involved in translation of genetic information into various proteins necessary for cellular processes.

Types and Functions of RNA:

There are three types of RNA:



- **Messenger RNA (mRNA):** It carries message of DNA from nucleus to cytoplasm for protein synthesis so called mRNA.
- **Transfer RNA (tRNA):** It transfer amino acids to ribosomes during protein synthesis hence called tRNA.
- **Ribosomal RNA (rRNA):** It is the component of ribosomes therefore, called rRNA. It combines with ribosomal protein to form units of ribosomes, which are factory of proteins.

